A Numerical Study of Turbulent Heat Transfer in Unconfined Impinging Slot Jets

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Introduction

Impinging jets have received considerable attention as they can provide large localized heat transfer for many applications such as annealing of metal, gas turbine heat transfer, and electronics cooling. Of particular interest are turbulent impinging jets which demonstrate superior heat transfer characteristics compared to laminar jets. This study presents a numerical study on the prediction of turbulent heat transfer in an unconfined single slot jet impingement on an isoflux flat plate. Governing equations are solved with a finite-volume approach in which the $k - \epsilon$ model and Reynolds-Stress Model (RSM) with enhanced wall functions are employed for the turbulence closure. Computations are performed at four jet Reynolds number based on the hydraulic diameter of the nozzle, $Re_j = 15000, 22500, 30000$ and 40000, where the non-dimensional nozzle-to-plate spacing, H/W varies between 2 and 8. Local Nusselt number, Nu predictions are compared with the available experimental data. Using the experimental data as a baseline, the objective is to compare the ability of $k - \epsilon$ and RSM closures to predict the stagnation, turning, and wall jet regions.

Turbulence modelling

Turbulent flow computations are performed by utilizing the $k - \epsilon$ model and RSM. Ensemble-averaged continuity and momentum equations can be written in tensor form:

$$\frac{\partial U_i}{\partial x_i} = 0 \tag{1}$$

$$\frac{\partial U_i}{\partial t} + \frac{\partial U_i U_j}{\partial x_j} = \frac{1}{\rho}\frac{\partial P}{\partial x_i} + \frac{\mu}{\rho}\Big[\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} - \frac{2}{3}\delta_{ij}\frac{\partial U_i}{\partial x_i}\Big] + \frac{1}{\rho}\frac{\partial \rho \prec u_i' u_j' \succ}{\partial x_j} \tag{2}$$

where U_i and P are the ensemble-averaged velocity components and pressure, respectively, u_i' denotes the fluctuating velocity components and $\prec\ \succ$ represents the ensemble average.

The turbulence kinetic energy, k and the dissipation rate, ϵ equations for the $k-\epsilon$ model are given by the following equations:

$$\frac{\partial k}{\partial t} + \frac{\partial k U_i}{\partial x_i} = \frac{1}{\rho}\frac{\partial}{\partial x_j}\Big[\Big(\mu + \frac{\mu_t}{\sigma_k}\Big)\frac{\partial k}{\partial x_j}\Big] - \frac{1}{\rho}(\rho \prec u_i' u_j' \succ \frac{\partial u_j}{\partial x_i}) - \epsilon \tag{3}$$

$$\frac{\partial \epsilon}{\partial t} + \frac{\partial \epsilon U_i}{\partial x_i} = \frac{1}{\rho}\frac{\partial}{\partial x_j}\Big[\Big(\mu + \frac{\mu_t}{\sigma_\epsilon}\Big)\frac{\partial \epsilon}{\partial x_j}\Big] + \frac{C_{1\epsilon}}{\rho}\frac{\epsilon}{k}(\rho \prec u_i' u_j' \succ \frac{\partial u_j}{\partial x_i}) + C_{2\epsilon}\frac{\epsilon^2}{k} \tag{4}$$

where μ_t is the turbulent viscosity determined from,

$$\mu_t = C_\mu \rho \frac{k^2}{\epsilon} \tag{5}$$

$C_{1\epsilon}$, $C_{2\epsilon}$, and C_μ are empirical constants and are taken as 1.44, 1.92 and 0.09, respectively. σ_k and σ_ϵ are the turbulent Prandtl numbers for k and ϵ and taken as 1.0 and 1.3, respectively.

The turbulent heat transport equation for $k-\epsilon$ model can be written as,

$$\frac{\partial(\rho E)}{\partial t} + \frac{\partial[U_i(\rho E + P)]}{\partial x_i} = \frac{\partial}{\partial x_j}\Big[k_e\frac{\partial E}{\partial x_j} + U_i\tau_{ij,e}\Big] \tag{6}$$

where E denotes the total energy per unit mass. k_e is the effective thermal conductivity determined by,

$$k_e = k_f + \frac{c_p \mu_t}{Pr_t} \tag{7}$$

where Pr_t is the turbulent Prandtl number and is taken as 0.85. The second term on the RHS of Equation (6) represents the viscous heating and can be shown in terms of effective viscosity and velocity gradients. Viscous heating is included in the present computations.

For RSM, the Reynolds' stress transport equation can be written as follows:

$$
\begin{aligned}
\frac{\partial(\rho \prec u_i' u_j' \succ)}{\partial t} \quad &+ \quad \frac{\partial(\rho u_k \prec u_i' u_j' \succ)}{\partial x_k} = \\
\frac{\partial}{\partial x_k}\Big[\rho \prec u_i' u_j' u_k' \succ \quad &+ \quad \prec P(\delta_{kj} u_i' + \delta_{ik} u_j'\Big] \\
+\frac{\partial}{\partial x_k}\Big[\mu \frac{\partial \prec u_i' u_j' \succ}{\partial x_k}\Big] \quad &- \quad \rho\Big[\prec u_i' u_k' \succ \frac{\partial U_j}{\partial x_k} + \prec u_j' u_k' \succ \frac{\partial U_i}{\partial x_k}\Big] \\
+\prec P\Big[\frac{\partial u_i'}{\partial x_j} + \frac{\partial u_j'}{\partial x_i}\Big] \succ \quad &- \quad 2\mu \prec \frac{\partial u_i'}{\partial x_j}\frac{\partial u_j'}{\partial x_i} \succ
\end{aligned}
\tag{8}
$$

where on the RHS, the first term represents turbulent diffusion, the second term represents molecular diffusion, the third term represents stress production, the fourth term represents pressure strain, and the last term represents dissipation. A linear pressure strain relation is employed in the RSM computations. The turbulent viscosity relation and the heat transport equation are identical to those of the $k - \epsilon$ model shown by Equations (5) and (6), respectively.

Near-wall modelling

Transport equations must be adjusted in the presence of walls to account for the behavior introduced by the no-slip condition. The characterization of the flow near the wall can be treated by utilizing wall functions or enhanced wall treatment. In the former, the link between the viscous sub-layer and turbulent flow is formed by wall functions, which can be written in equilibrium and non-equilibrium forms. Non-equilibrium effects are partially considered in the non-equilibrium wall functions. If the low-Reynolds number effects are significant, it is known that wall functions, either equilibrium or non-equilibrium forms, cannot resolve the near-wall flow reasonably since the empirical and semi-empirical relations are commonly determined from high Reynolds number turbulent flows. The wall function approach is also limited in the sense that it is not responsive to rapid strain and turbulent length scale changes, therefore their use is not suitable in the case of jet impingement.

The enhanced wall treatment approach solves the governing equations all the way to the wall. Enhanced wall treatment produces significantly better results compared to wall functions in flows under consideration. However, it is computationally more expensive. Additional computational expense comes from the requirement of fine near-wall meshing required to resolve the viscous sub-layer adjacent to the wall.

A turbulent Reynolds number, Re_y, can be defined as,

$$Re_y = \frac{\rho y \sqrt{k}}{\mu} \tag{9}$$

where y is the distance between the wall and the center of the computational cell under interest. Molecular viscosity effects are significant and Equation (5) is not valid in the region of $Re_y < 200$. In this region, the flow field is solved by the enhanced wall functions which are obtained by blending of the enhanced turbulent wall law and laminar wall law. The meshing at the near-wall region should be fine enough to simulate the fluid flow correctly. In the present study, there are at least 10 near-wall region nodes, the first of which lies in $1 < y^+ < 5$. If $Re_y > 200$, the transport equations for fully-turbulent flow given in the previous section are utilized.

While not shown here, it should be mentioned that the computations are repeated with different versions of the $k - \epsilon$ model. No significant difference can be observed among them. The same behavior was reported for axi-symmetric jets.

Results and Discussions

The computational domain is illustrated in Figure 1. A general purpose finite volume CFD program is utilized for the computations. A non-uniform mesh is constructed with 4 node rectangular cells with finer meshing near the walls. Diffusive terms are central-differenced and convective terms are discretized with the QUICK scheme. For pressure-velocity coupling, the SIMPLEC algorithm is utilized. Isoflux boundary conditions are imposed on the walls, where constant pressure boundary conditions are imposed for both the upper and the side boundaries. Nozzle-exit velocity and turbulence profiles for the nozzle are obtained from the fully simulated flows of the actual nozzle geometry.

Results from the standard $k - \epsilon$ model with standard wall functions are compared with those of enhanced wall functions with two different turbulence models at $Re_j = 40000$, $H/W = 6$ in Figure 2. There is no appreciable difference between equilibrium and non-equilibrium standard wall functions; the results shown are obtained with the former. As illustrated in Figure 2, standard wall functions significantly over-estimate Nu around the stagnation point. In this particular case, Nu_{st} is over-estimated by 40% while the difference is as high as 90% for different nozzle-to-plate spacings and jet velocities. Similar results are presented for circular jets, in which this difference was attributed to the erroneous physics of the $k-\epsilon$ model. Even though the computational expense decreases dramatically, the $k-\epsilon$ model with standard wall functions demonstrates an unrealistic behavior and should be avoided in the case of turbulent stagnation flows.

Figures 3 and 4 demonstrate Nu distribution at $Re_j = 15000, 40000$ and $H/W = 2$. RSM results are in good agreement with the experimental data around the stagnation region up to $x/W = 5$. Results from the $k-\epsilon$ model not only over-predict the experimental data but, more importantly, demonstrate an incorrect stagnation point area behavior. Experimental data demonstrates a secondary maximum for the given height and Re_j range, which is even more pronounced with increasing Re_j. In RSM computations, although the shape of Nu curve becomes less steep in that region, the secondary maximum is not well-defined except for high Re_j. Both RSM and $k-\epsilon$ model adequately resolve the wall jet behavior further downstream.

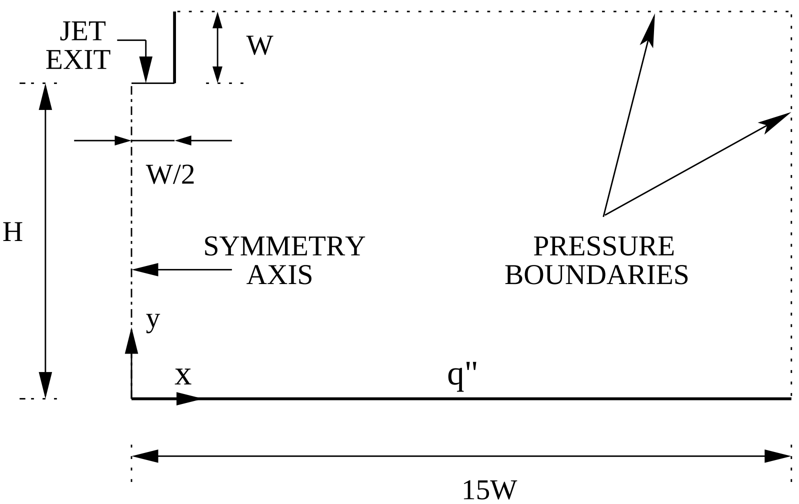

Figure 1: Computational domain.

The secondary maximum far from the stagnation region shown by the experimental data is somewhat captured by RSM simulations, however this is not the case for $k-\epsilon$ model. If the turbulence kinetic energy profiles are investigated, those from RSM simulations demonstrate a clear, continuous increase in the turbulence kinetic energy up to the secondary maximum. In contrast, those from the $k-\epsilon$ model show a high level of turbulence kinetic energy around the stagnation region, and a rapid dissipation in the wall jet direction. As mentioned previously, in the open literature, the secondary maxima behavior is commonly attributed to *relaminarization*. However, in the case of radial jets, computations performed for radial jets showed a radial increase in turbulent kinetic energy in agreement with the corresponding experimental work. For slot jets, a similar behavior to radial jets is observed based on the results from RSM.

Increasing nozzle-to-plate spacing, the steep secondary maxima are significantly flattened. Figures 5 and 6 show local Nusselt number behavior at $Re_j = 15000, 40000$ and $H/W = 6$. Similar to that of $H/W = 2$ case, the $k-\epsilon$ model predicts a secondary peak close to the stagnation point. RSM results show excellent agreement with the experimental data up to $x/W \approx 5$ and are able to capture wall jet-behavior further downstream. It was hypothesized that in the region of secondary maxima, RSM is not able to capture the local physics. Inspection of the turbulence profiles from the RSM simulations at $H/W = 6$ show that the turbulent kinetic energy in the vicinity of secondary maxima is reduced compared to the stagnation area, however it would appear that perhaps the rate of turbulent kinetic energy production is too low, causing an under-prediction of heat transfer.

Variation of Nu_{st} with Re_j is investigated at two different nozzle-to-plate spacings, $H/W = 2$ and $H/W = 6$ as shown in Figures 7 and 8. RSM simulations demonstrate excellent agreement with the experimental data at both spacings. At $H/W = 2$, the best fit to the data gives a dependence of $\sim Re_j^{0.51}$, where RSM estimates a dependence of $\sim Re_j^{0.57}$. At $H/W = 6$, the best fit to the data gives a dependence of $\sim Re_j^{0.68}$, where RSM estimates a dependence of $\sim Re_j^{0.62}$. The dependence of Nu_{st} on Re_j cannot be determined correctly by the $k-\epsilon$ model.

Effect of velocity profiles at nozzle exit

It was shown that the effects of nozzle-exit profiles on Nu distribution are significant around the stagnation area in the case of turbulent radial jets. A similar investigation is presented here for turbulent slot jets.

Table 1: Summary of nozzle-exit profiles

Case	CL vel. ratio	Tu. int.
1: actual	1.07	2.9%
2: uniform	1.00	2.9%
3: uniform	1.00	5.0%
4: $\parallel$ plates	1.11	3.0%

Computations are performed for various nozzle-exit profiles, where $H/W = 6$. Four different cases are investigated, and shown at Table 1. In all these cases, mean velocity at the nozzle exit is equal which corresponds to a $Re_j = 22500$. In the first case, a uniform flow is assumed at the nozzle inlet with a turbulence intensity of 0.01% and the flow field at the actual nozzle geometry is solved so that the turbulence and velocity profiles are obtained at the nozzle exit. In the second and third cases, the velocity profile at the nozzle exit is assumed to be uniform with different turbulence intensities, namely 2.9% and 5%, respectively. In the fourth case, the nozzle-exit profile is obtained from the fully developed flow between parallel plates with the velocity profile entering through the plates uniform and turbulence intensity being 0.01%.

Figure 9 demonstrates the Nu distribution for all the cases in the range of $0 \leq x/W \leq 5$. For the first case, computations are performed with both $k - \epsilon$ model and RSM where the difference between Nu_{st} is around 10%. While the $k - \epsilon$ model determines the maximum at $x/W = 0.25$, RSM computations demonstrates that maximum Nu is obtained at the stagnation point. Results from RSM are in excellent agreement with the experimental data. Comparing cases 1 and 4, $k - \epsilon$ model give almost identical results for both. However, if a uniform velocity profile is assumed at the nozzle exit as in cases 2 and 3, regardless of the turbulence intensity, near stagnation point Nu behavior cannot be captured either qualitatively or quantitatively. Nu_{st} values for cases 2 and 3 are within 2% of each other, while the difference between experimental data or RSM is more than 20%.

Nu behavior with three different nozzle-exit profiles at $Re_j = 23000$ and a low nozzle-plate spacing($H/D = 2$) for radial jets was compared in the literature. The difference in Nu distributions with different nozzle-exit profiles was attributed to low nozzle-plate spacing, as the free jet is not fully-developed. However, the nozzle-plate spacing may not be the primary reason to observe the difference in Nu distribution, as the computations for slot jets presented here are performed at a moderate nozzle-plate spacing($H/W = 6$) and still show a considerable dependence of Nu distribution on the nozzle-exit profiles.

Summary

Results from the $k - \epsilon$ model with standard wall functions (either equilibrium or non-equilibrium) over-estimate the experimental data significantly in the close vicinity of stagnation point area. RSM computations with enhanced wall functions demonstrated good agreement with the experimental data at given H/W and Re_j range, especially at high H/W. The $k - \epsilon$ model with enhanced wall functions determines an incorrect stagnation point area behavior. RSM can capture Nu_{st} vs Re_j characteristics, while $k - \epsilon$ model cannot. Optimal H/W based on Nu_{st} is also determined, RSM yields a value around 7, where $H/W \approx 8$ is determined by the experiments.

Similar to that of axi-symmetric jets, it was shown that velocity and turbulence profiles at the nozzle exit are quite significant in determining the Nusselt number, both quantitatively and qualitatively. Uniform profiles considerably underestimate Nu values.

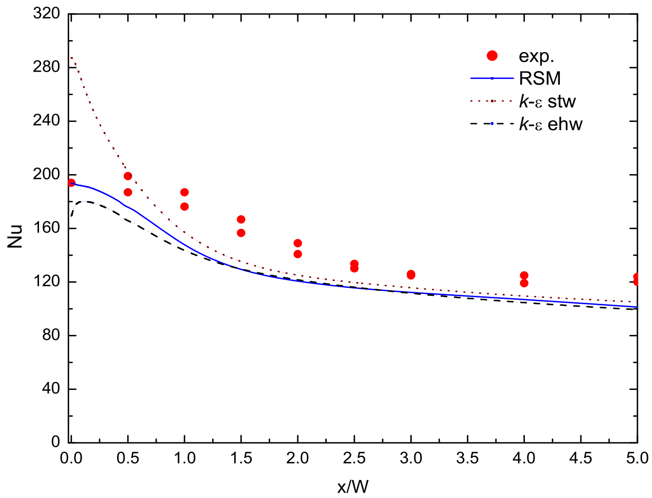

Figure 2: Comparison of standard wall functions with enhanced wall functions, Nu vs. x/W at $Re_j = 40000$ and $H/W = 6$.

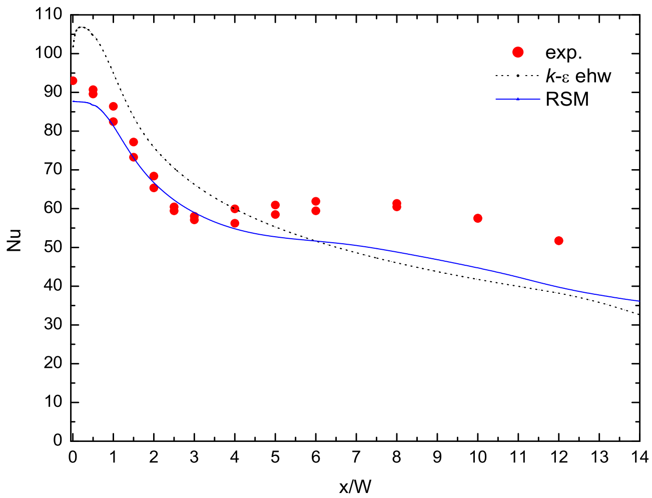

Figure 3: Variation of Nu vs. x/W at $Re_j = 15000$ and $H/W = 2$.

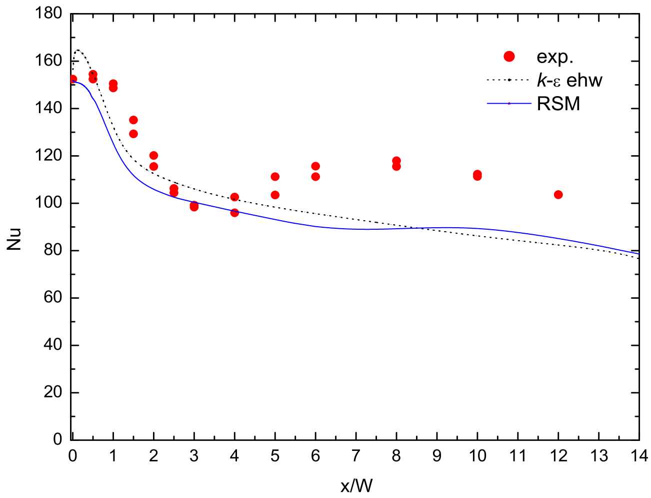

Figure 4: Variation of Nu vs. x/W at $Re_j = 40000$ and $H/W = 2$.

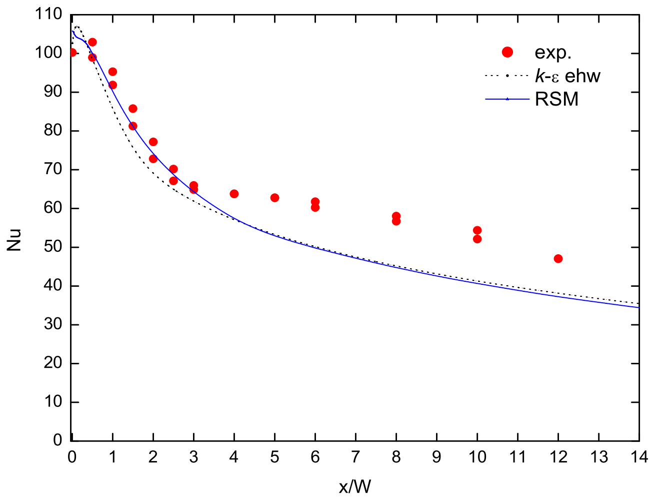

Figure 5: Variation of Nu vs. x/W at $Re_j = 15000$ and $H/W = 6$.

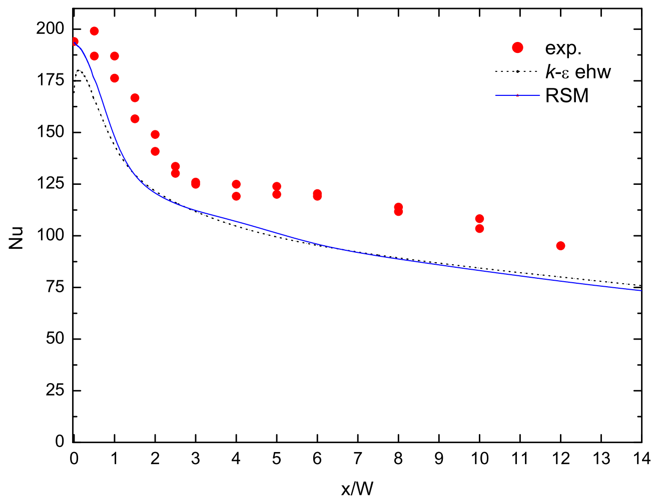

Figure 6: Variation of Nu vs. x/W at $Re_j = 40000$ and $H/W = 6$.

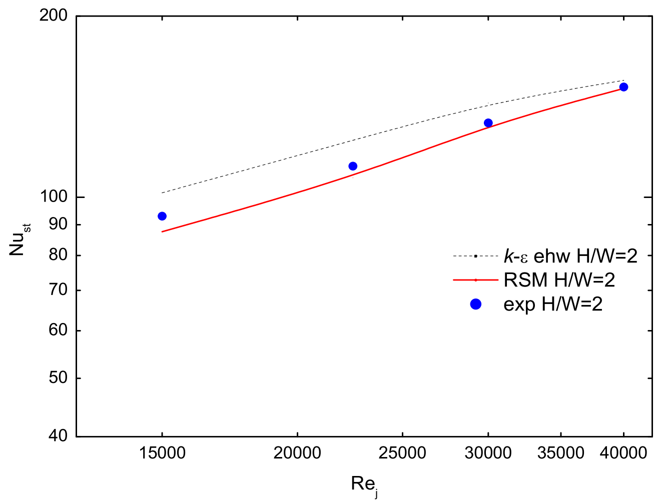

Figure 7: Variation of Nu_{st} vs. Re_j at $H/W = 2$.

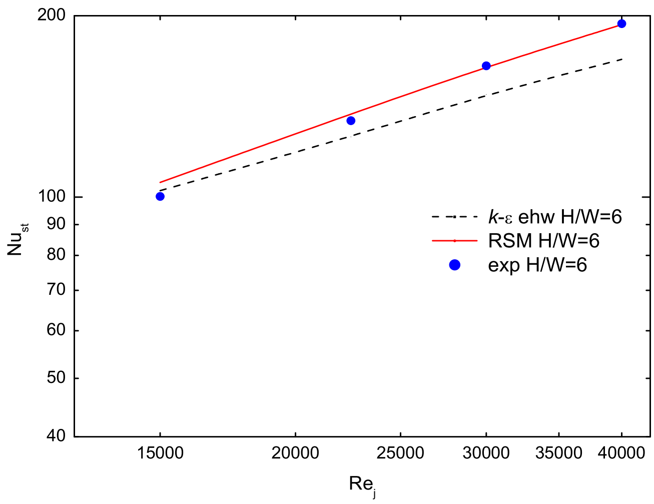

Figure 8: Variation of Nu_{st} vs. Re_j at $H/W = 6$.

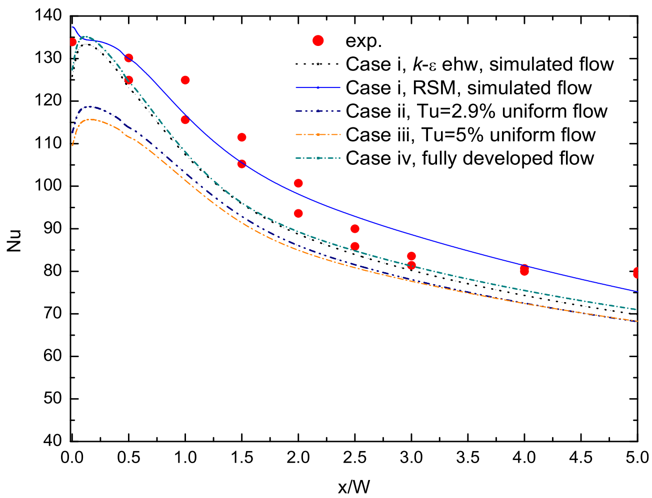

Figure 9: Nu vs. x/W for the cases given in Table1 at $Re_j = 22500$ and $H/W = 6$.